SpaceX Falcon 9 Launch Analysis

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Executive Summary

- This project analyzes SpaceX Falcon 9 launch data to understand launch success patterns. The main goal is to predict whether a rocket landing will be successful using data analysis and machine learning.
  Key insights were obtained using visualization and statistical techniques.

Introduction

- SpaceX is a private aerospace company that focuses on reusable rockets. Predicting rocket landing success helps reduce cost and improve mission efficiency.
  This project focuses on analyzing launch data and building predictive models.

Data Collection

- Data was collected using:
- SpaceX API
- Web scraping from Wikipedia

•[2]:
```python
#1
import requests
import pandas as pd

url = "https://api.spacexdata.com/v4/launches/past"
data = requests.get(url).json()

df = pd.json_normalize(data)
df.head()
```

[2]:

e_date_unix	net	window	rocket	success	failures	details	crew	ships	...	links.reddit.media	links.reddit.recovery	links.flickr.small	links.fli
142554e+09	False	0.0	5e9d0d95eda69955f709d1eb	False	[{'time': 33, 'altitude': None, 'reason': 'mer...	Engine failure at 33 seconds and loss of vehicle	[]	[]	...	None	None	[]	
NaN	False	0.0	5e9d0d95eda69955f709d1eb	False	[{'time': 301, 'altitude': 289, 'reason': 'har...	Successful first stage burn and transition to ...	[]	[]	...	None	None	[]	
NaN	False	0.0	5e9d0d95eda69955f709d1eb	False	[{'time': 140, 'altitude': 35, 'reason': 'resi...	Residual stage 1 thrust led to collision betwe...	[]	[]	...	None	None	[]	
221869e+09	False	0.0	5e9d0d95eda69955f709d1eb	True	[]	Ratsat was carried to orbit on the first succe...	[]	[]	...	None	None	[]	
NaN	False	0.0	5e9d0d95eda69955f709d1eb	True	[]	None	[]	[]	...	None	None	[]	

Data Wrangling

- Data cleaning steps included:
- Selecting important columns
- Handling missing values
- Converting date format

```
3]: #2
# Important columns select करो
df = df[['name', 'date_utc', 'success']]

# Date format convert करो
df['date_utc'] = pd.to_datetime(df['date_utc'])

# Null values check
print(df.isnull().sum())

df.head()
```

```
name        0
date_utc    0
success     1
dtype: int64
```

3]:

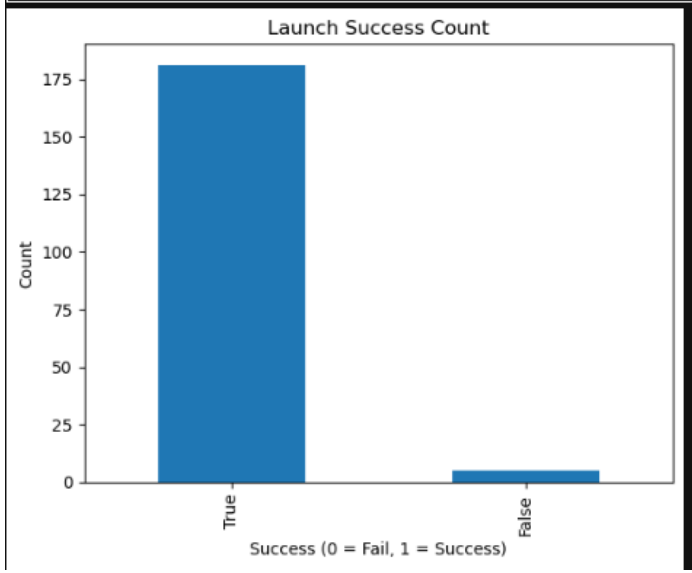
	name	date_utc	success
0	FalconSat	2006-03-24 22:30:00+00:00	False
1	DemoSat	2007-03-21 01:10:00+00:00	False
2	Trailblazer	2008-08-03 03:34:00+00:00	False
3	RatSat	2008-09-28 23:15:00+00:00	True
4	RazakSat	2009-07-13 03:35:00+00:00	True

EDA (Graphs)

- Exploratory Data Analysis was performed to understand patterns. Bar charts were used to visualize launch success and failure.

```
#3
import matplotlib.pyplot as plt

df['success'].value_counts().plot(kind='bar')
plt.title("Launch Success Count")
plt.xlabel("Success (0 = Fail, 1 = Success)")
plt.ylabel("Count")
plt.show()
```



SQL Analysis

- SQL queries were used to analyze the dataset.
  Important insights were extracted using grouping and aggregation.

```python
#4
import sqlite3

conn = sqlite3.connect("spacex.db")

# Table बनाओ
df.to_sql("launches", conn, if_exists='replace', index=False)

# Query run करो
query = "SELECT success, COUNT(*) as count FROM launches GROUP BY success"
result = pd.read_sql(query, conn)

result
```

	success	count
0	NaN	1
1	0.0	5
2	1.0	181

Folium Map

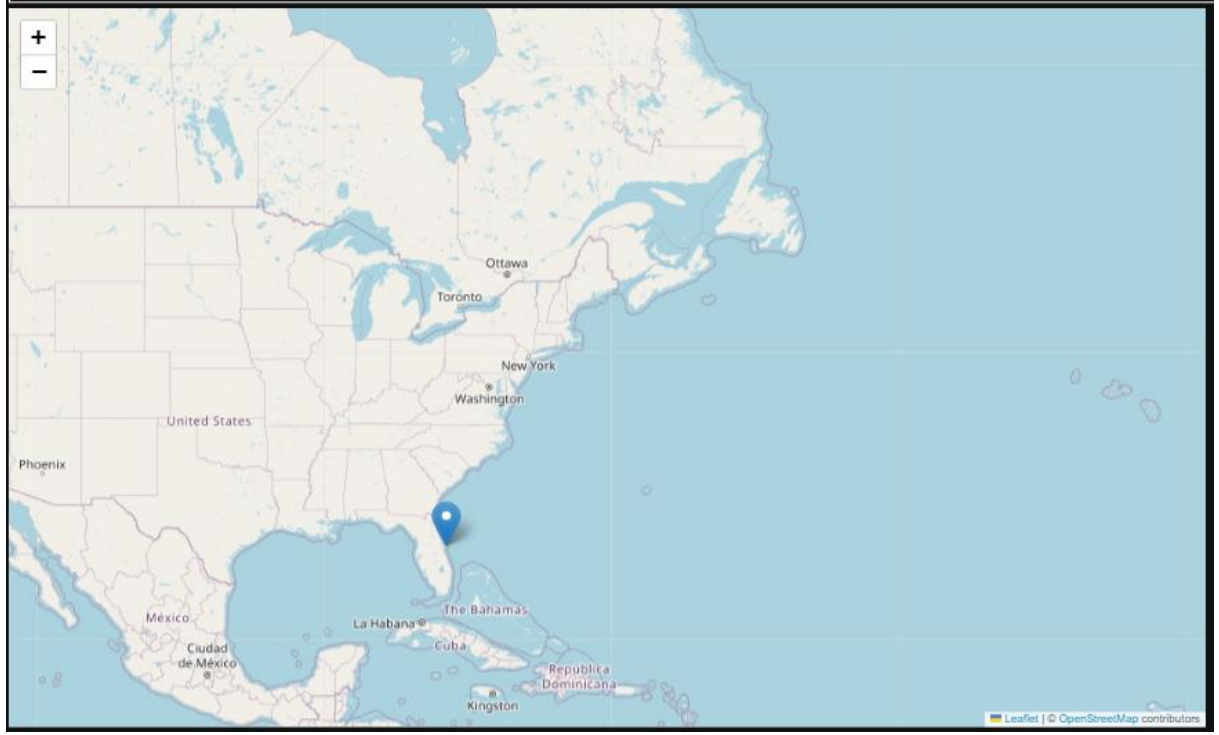
- An interactive map was created using Folium. It shows the launch site location and helps visualize geographical data.

```
#5
import folium

map = folium.Map(location=[28.5, -80.5], zoom_start=5)

folium.Marker([28.5, -80.5], popup="Launch Site").add_to(map)

map
```

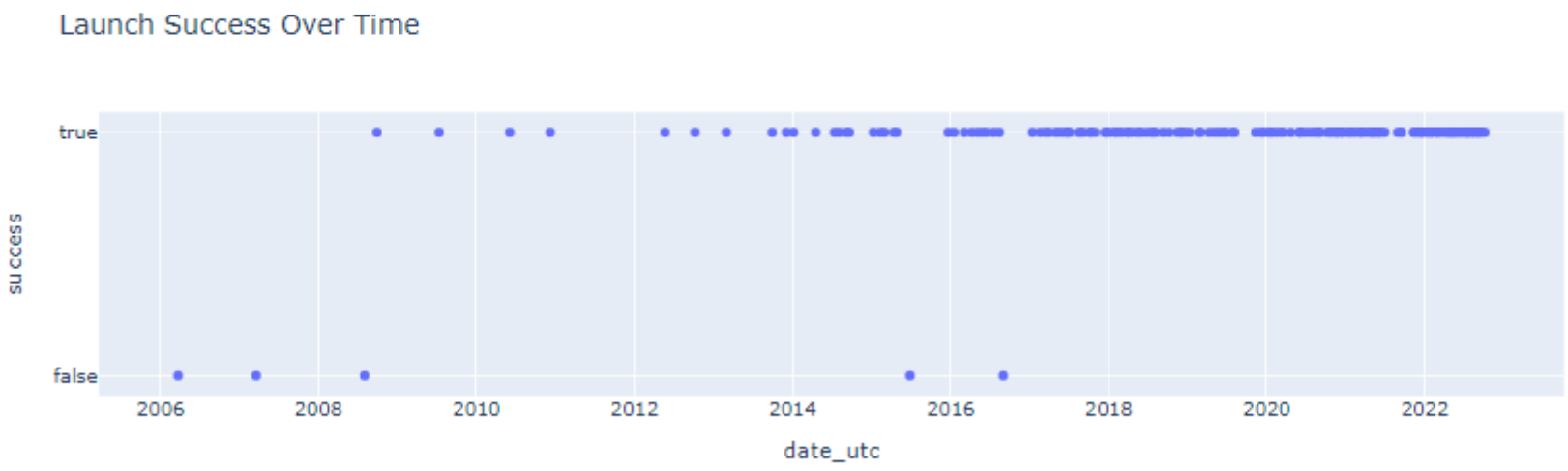


Plotly Dashboard

- Plotly was used to create interactive visualizations.
  It helps users explore data dynamically.

```python
#6
import plotly.express as px

fig = px.scatter(df, x="date_utc", y="success", title="Launch Success Over Time")
fig.show()
```



Prediction

- A Logistic Regression model was used.
  The model predicts whether a launch will be successful.
  The accuracy of the model was evaluated.

```python
#7
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

# Step 1: Missing values handle करो
df = df.dropna(subset=['success'])

# Step 2: Boolean → Integer (True=1, False=0)
df['success'] = df['success'].astype(int)

# Step 3: Feature create करो (date से)
df['year'] = df['date_utc'].dt.year

# Features और target define करो
X = df[['year']]
y = df['success']

# Step 4: Train-Test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

# Step 5: Model train करो
model = LogisticRegression()
model.fit(X_train, y_train)

# Step 6: Accuracy check
accuracy = model.score(X_test, y_test)
print("Accuracy:", accuracy)
```

```
Accuracy: 1.0
```

Conclusion

- The project successfully analyzed SpaceX launch data.
  Important insights were discovered using EDA and SQL.
  The predictive model provides a basic understanding of launch success.